
A Survey on Vision-Language-Action Models: Foundations, Architectures, and Open Challenges

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Abstract

Vision-Language-Action (VLA) models represent a paradigm shift in embodied AI, transitioning from specialized robotic controllers to generalist foundation models capable of reasoning and actuation. This survey provides a comprehensive review of recent advancements in VLAs. We propose a taxonomy based on model architecture, learning paradigms, and enhanced capabilities. We analyze state-of-the-art methods ranging from transformer-based policies to diffusion world models, discuss the critical role of large-scale datasets, and highlight open challenges in the field.

1 Introduction

1.1 Background

The recent convergence of large language models (LLMs) and computer vision has led to the emergence of Vision-Language Models (VLMs), which excel at grounding linguistic understanding in visual perception. Building on this synergy, the research community has extended these models into the realm of physical control, giving rise to Vision-Language-Action (VLA) models. Unlike traditional robotic pipelines that rely on modular perception, planning, and control stages, VLAs process multimodal inputs—including images, depth, proprioception, and natural language instructions—to directly generate low-level control actions, enabling more flexible and generalizable behaviors.

1.2 Motivation

Traditional robotic learning approaches, such as task-specific imitation learning or reinforcement learning, often struggle with generalization to unseen tasks, objects, and environments. VLAs aim to overcome these limitations by leveraging the semantic knowledge embedded in web-scale text and image datasets. By combining language grounding, visual perception, and action learning, VLA models hold the promise of performing long-horizon, multi-step tasks in complex, open-world environments, while reducing the need for extensive task-specific demonstrations.

1.3 Scope and Structure

This survey introduces the problem definition of VLAs and mainly focuses on three key aspects of VLA research: (i) end-to-end foundation models trained on large-scale multimodal data, (ii) reinforcement learning adaptations for policy refinement and self-improvement, and (iii) capability extensions including 3D reasoning, memory augmentation, and latent action representations. The remainder of this paper is structured as follows: Section 2 formalizes the VLA problem and presents a taxonomy of existing approaches. Section 3 reviews the data foundations supporting VLA learning. Section 4 provides a detailed survey of state-of-the-art methods. Section 5 outlines current challenges and open research issues. Finally, Section 6 summarizes key insights and highlights future directions.

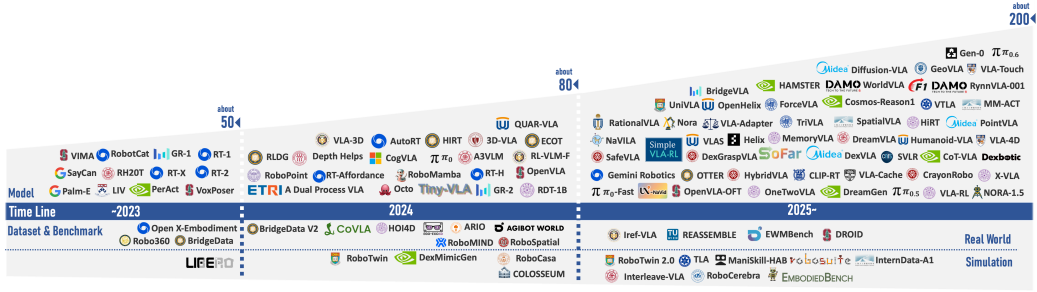


Figure 1: The timeline of VLA works in recent years. (This figure comes from Xu et al. [2025])

2 Problem Definition

2.1 Core Problem Formulation

Vision–Language–Action (VLA) models aim to endow embodied agents with the ability to execute complex tasks specified in natural language by leveraging multimodal sensory inputs and producing continuous control actions. Formally, a VLA model can be represented as a policy function π that operates in a sequential decision-making setting. At each time step t , the policy takes as input the current observation O_t , a language instruction L , and optionally the previous action a_{t-1} , and outputs the next action a_t :

$$a_t = \pi(a_{t-1}, O_t, L). \quad (1)$$

Here, the observation O_t typically consists of high-dimensional visual inputs (e.g., RGB or RGB-D images) and proprioceptive signals (e.g., joint positions, velocities, or force feedback), capturing both the external environment and the agent’s internal state. The language instruction L encodes high-level task intent, often expressed in abstract, compositional, and temporally underspecified natural language. The action a_t lies in a continuous control space, such as joint torques, end-effector poses, or velocity commands, depending on the embodiment of the agent.

The core challenge of the VLA problem lies in bridging the substantial representational and semantic gap between language and action. Specifically, language instructions convey high-level, symbolic, and goal-oriented semantics, while actions correspond to low-level, continuous motor commands that must respect physical constraints, system dynamics, and real-time feedback. Effective VLA models must therefore learn a grounded alignment that maps linguistic concepts (e.g., objects, relations, and temporal goals) onto perceptual features and translates them into temporally coherent motor behaviors. This alignment problem is further complicated by partial observability, long-horizon dependencies, and the need for generalization across diverse tasks, environments, and linguistic expressions.

2.2 Taxonomy

To align with the organization of state-of-the-art VLA methods, we propose a taxonomy that categorizes existing approaches according to how they *parameterize*, *train*, and *extend* end-to-end Vision–Language–Action policies. Rather than treating architectural and learning choices as fully orthogonal axes, this taxonomy reflects the dominant design patterns observed in contemporary VLA systems and directly mirrors the structure of our SOTA review.

1. **Policy Parameterization: Action Representation and Generation.** A primary distinction among VLA models lies in how actions are represented and generated within the policy. *Action-token-based* approaches embed actions as discrete or quantized tokens within an autoregressive Transformer, enabling unified sequence modeling over vision, language, and action. This design tightly couples perception, instruction following, and control, and is particularly amenable to large-scale pretraining via behavior cloning. In contrast, *action-expert-based* approaches decouple high-level multimodal reasoning from low-level control by delegating action generation to specialized modules, such as inverse kinematics solvers,

trajectory generators, or learned motor primitives. This separation often improves control fidelity and facilitates transfer across robot embodiments, at the cost of weaker end-to-end coupling.

2. **Learning Paradigms: From Imitation to Interactive Optimization.** State-of-the-art VLA systems differ substantially in how policies are optimized. *Pure imitation learning* methods rely exclusively on large-scale demonstration datasets, offering stability and scalability but limited capacity for self-correction beyond the data distribution. *Reinforcement learning-augmented* approaches incorporate reward-driven optimization to refine pretrained policies through interaction, enabling long-horizon credit assignment, robustness to compounding errors, and adaptation to novel tasks or environments. Importantly, RL in VLA systems is typically not applied from scratch, but used as a refinement mechanism on top of pretrained VLA backbones, whether by fine-tuning action tokens, updating action experts, or adjusting high-level decision variables.
3. **System-Level Extensions for Long-Horizon and Real-World Tasks.** Beyond core policy modeling and training, many SOTA VLA systems integrate additional components to address challenges inherent to real-world deployment. *Hierarchical organization* introduces temporal abstraction by separating high-level task planning from low-level control, often combining language-conditioned planners with continuous controllers. *Explicit spatial grounding* mechanisms incorporate 3D representations, such as point clouds, scene graphs, or neural fields, to improve geometric reasoning and physical interaction. *Memory and state abstraction* modules enable information retention across time or episodes, supporting partial observability and long-horizon instruction execution.

This taxonomy emphasizes how modern VLA systems are structured and trained in practice, highlighting recurring design patterns across state-of-the-art methods. By organizing prior work along these dimensions, it provides a coherent framework for understanding how architectural choices, learning paradigms, and system-level extensions jointly shape the capabilities and limitations of contemporary VLA agents.

3 Data Foundations

The performance and generalization capability of Vision–Language–Action (VLA) models are critically dependent on the scale, diversity, and structure of their training data. In contrast to vision–language models that focus solely on perceptual alignment, VLA systems must learn grounded correspondences among natural language instructions, high-dimensional sensory observations, and low-level continuous motor actions. This requirement renders data collection substantially more expensive and heterogeneous, and has made data design a central driver of recent progress in the field.

- **Robotic Demonstrations.** Robot demonstration datasets provide direct supervision over the action space and constitute the primary data source for imitation-based VLA training. The **Open X-Embodiment** dataset O’Neill et al. [2024] represents a significant step toward scalable data collection by aggregating demonstrations across diverse robot embodiments, tasks, and sensor configurations. By unifying heterogeneous platforms under a shared representation, it enables cross-embodiment transfer and supports the training of generalist policies such as RT-X O’Neill et al. [2024]. This shift from single-robot, task-specific datasets to multi-robot pretraining improves scalability, while simultaneously introducing challenges related to distributional diversity and action-space alignment.
- **Egocentric Video.** Large-scale egocentric human video datasets provide complementary supervision for learning interaction semantics, object affordances, and long-horizon task structure without requiring explicit robot actions. Datasets such as **Ego4D** Grauman et al. [2022] capture first-person human activities across diverse real-world scenarios, offering temporally dense visual streams aligned with natural language annotations. In VLA systems, such data is typically leveraged for representation pretraining, enabling models to acquire semantic and spatial priors that can be transferred to robotic settings via fine-tuning or adaptation. Despite the absence of embodiment-specific action labels, egocentric data plays a crucial role in scaling visual–language understanding and reducing reliance on costly robotic demonstrations.

- **Synthetic and Simulated Data.** Synthetic data generated in simulation environments provides a scalable alternative for collecting aligned vision, language, and action signals. Simulation platforms allow precise control over scene configurations, object properties, task variations, and reward signals, enabling large-scale data generation at relatively low cost. Such data is commonly used to train VLA models on long-horizon task structure, spatial reasoning, and exploration strategies before transfer to real-world settings. However, discrepancies between simulated and real-world dynamics, visual appearance, and contact physics introduce a sim-to-real gap, necessitating careful domain randomization, representation learning, or real-world fine-tuning.

Table 1: Representative Datasets for Vision–Language–Action (VLA) Models

Dataset	Real Robot	Language	Vision	Action	Synthetic
Open X-Embodiment	✓	✓	✓	✓	
DROID	✓	✓	✓	✓	
RoboNet	✓		✓	✓	
BridgeData (V1/V2)	✓		✓	✓	
Language Table	✓	✓	✓	✓	
Ego4D		✓	✓		
EPIC-KITCHENS		✓	✓		
ALFRED (AI2-THOR)		✓	✓	✓	✓
RLBench			✓	✓	✓
Habitat / R2R		✓	✓		✓

Together, these data modalities form a hybrid data foundation for VLA models, combining the grounding fidelity of robotic demonstrations, the semantic richness of human-centric videos, and the scalability of synthetic data. Modern VLA systems increasingly rely on integrating these complementary sources to balance data efficiency, generalization, and embodiment specificity.

4 State-of-the-Art Methods

4.1 End-to-End Foundation Models

End-to-end foundation models dominate contemporary VLA research by learning a single policy that maps multimodal observations and natural language instructions directly to low-level control actions, without explicit task decomposition or symbolic planning. These models are typically trained via behavior cloning on large-scale robot demonstration datasets, leveraging their representational capacity to handle task diversity, embodiment variation, and linguistic compositionality.

A key feature of end-to-end foundation models is scale, both in terms of model capacity and data diversity. Pretraining or fine-tuning on heterogeneous robot datasets—often combined with large vision–language corpora—enables these models to function as generalist policies capable of zero-shot or few-shot transfer across tasks, objects, and environments.

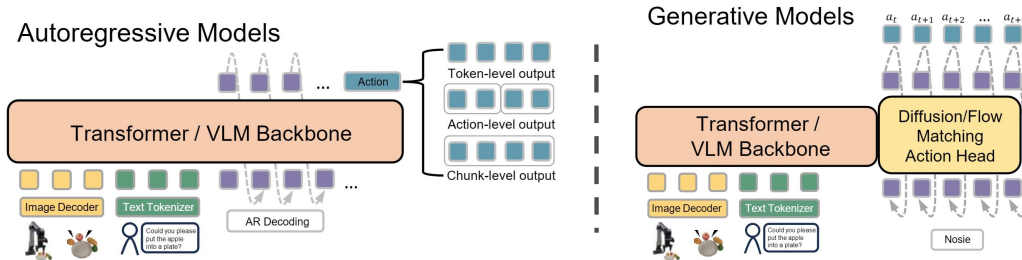


Figure 2: Two representative architectural paradigms for VLA models: end-to-end foundation models and hierarchical reasoning-based models.

4.1.1 Actions as Tokens

Early VLA systems extend Transformers by treating robot actions as part of a unified token sequence. Continuous control signals are discretized or embedded into tokens, allowing actions to be generated autoregressively alongside visual and linguistic tokens. This design facilitates tight multimodal alignment and enables direct reuse of pretrained vision–language models.

- **RT-1** Brohan et al. [2022] and **RT-2** Brohan et al. [2023] show that co-fine-tuning large vision–language models on robot demonstration data transfers semantic knowledge from web-scale text and images into embodied actions. By unifying language, vision, and action tokens, RT-2 generalizes to novel instructions and unseen objects, demonstrating the effectiveness of internet-scale pretraining for robotic manipulation.
- **OpenVLA** Kim et al. [2024] provides an open-source reference for Transformer-based VLA, emphasizing reproducibility. Released pretrained checkpoints and optimized training pipelines facilitate the study of data scaling, architectural choices, and evaluation protocols.
- **GR00T** Bjorck et al. [2025] extends the paradigm to humanoid robotics with high-dimensional action spaces and complex kinematics. By focusing on whole-body coordination, GR00T shows that token-based Transformers can scale beyond tabletop manipulation to general embodied agents.

Despite their success, token-based policies face challenges in action discretization, temporal smoothness, and representing multimodal continuous distributions, motivating alternative generative formulations.

4.1.2 Diffusion- and Flow-Based Models

To capture the continuity and multimodality of action spaces, diffusion- or flow-based generative models parameterize policies as conditional continuous distributions. Actions are sampled through iterative denoising or continuous transformations, which suits tasks with multiple valid solutions and smooth control requirements.

- π_0 and $\pi_{0.5}$ Black et al. [2024] introduce a flow-matching objective for general-purpose robot control, improving training stability and efficiency. Conditioned on observations and instructions, these models produce coherent, smooth trajectories across diverse manipulation tasks.
- **RDT** Liu et al. [2024a] and **Octo** Octo Model Team et al. [2024] employ diffusion Transformers as action experts, decoupling high-level multimodal reasoning from low-level control. This allows modeling complex action distributions while leveraging Transformer backbones.

Diffusion- and flow-based models complement token-based Transformers by naturally handling continuous actions, though inference costs are higher.

4.1.3 Alternative Architectures

Beyond Transformers and diffusion models, other sequence modeling approaches have been explored.

- **RoboMamba** Liu et al. [2024b] uses the state-space model **Mamba** Gu and Dao [2024] as a sequence backbone for vision–language–action modeling. Mamba’s linear-time processing and long-range dependency handling provide an efficient alternative for long-horizon robotic control.

4.2 VLA with Reinforcement Learning

Reinforcement learning (RL) complements large-scale foundation models by enabling interaction-driven optimization and robustness. In current VLA systems, RL is rarely applied end-to-end; instead, it is integrated modularly, with its role shaped by how actions are represented.

Transformer-based RL methods, such as Q-Transformer Chebotar et al. [2023], show that sequence models can incorporate reward-aware objectives via autoregressive value prediction. While not VLA systems themselves, these methods provide foundations for RL integration into VLA architectures.

The effectiveness of RL depends on action representation. In token-based models (e.g., RT-2, OpenVLA), actions are discrete tokens, making RL challenging due to large action spaces, sparse rewards, and unstable exploration. In contrast, action expert-based systems decouple high-level reasoning from low-level control: the VLA model outputs abstract action representations or goals, and a specialized expert generates continuous commands. RL can be applied naturally at the control level, where rewards are well-defined and exploration is constrained. A representative example is $\pi_{0.6}^*$ Amin et al. [2025], which updates its action expert offline using interaction data while keeping the high-level model fixed.

Modular, hierarchical integration of RL—focused on action experts while high-level policies rely on large-scale imitation—provides a practical path toward robust, self-improving VLA agents in long-horizon, open-world settings.

4.3 Hierarchical and Reasoning-based VLAs

Hierarchical VLA approaches decompose vision–language–action learning into multiple reasoning and control levels, separating high-level planning from low-level execution. This improves interpretability, sample efficiency, and generalization to long-horizon, compositional tasks. High-level modules can incorporate symbolic reasoning, skill abstractions, or pretrained language models, while learned controllers ensure robust execution.

- **SayCan** Ahn et al. [2022] uses LLMs for task planning (“Say”) and grounds plans with learned affordances over robot skills (“Can”). This combination enables assessing plan feasibility while maintaining flexibility, achieving strong performance on long-horizon manipulation.
- **FiS-VLA** Chen et al. [2025] integrates fast and slow reasoning systems within a single model, restructuring final Transformer layers of a slow System 2 into an efficient execution module (System 1). This fusion of deliberative and reactive control improves efficiency and adaptability.
- **ECoT** Zawalski et al. [2024] introduces explicit chain-of-thought reasoning, generating intermediate reasoning steps before action execution. Structured traces enhance transparency and improve performance on tasks requiring multi-step logic and long-term dependencies.

Hierarchical and reasoning-based architectures offer modular, interpretable VLA systems capable of handling complex, long-horizon tasks in open-world environments.

4.4 Enhanced Representations: 3D, Memory, and Action

Recent research incorporates richer, structured representations to capture spatial, temporal, and action-related context, addressing the limitations of conventional 2D end-to-end models in long-horizon tasks.

- **3D Intelligence: 3D-VLA** Zhen et al. [2024] uses depth and point clouds for reasoning about object geometry, spatial relations, and interactions, enabling predictive planning and physics-aware manipulation. **WorldVLA** Cen et al. [2025] enhances the **Chameleon** backbone Team [2024] by jointly predicting next images and actions conditioned on current observations, integrating world-model knowledge into policy learning.
- **Memory Augmentation: MemoryVLA** Shi et al. [2025] stores task-relevant context over extended horizons, supporting sequential manipulations, multi-room navigation, and complex assembly while reducing compounding errors.
- **Action Representation: Latent Action Pretraining** Ye et al. [2024] learns low-dimensional embeddings from human video datasets, aligning high-dimensional robot proprioception with demonstrations. Latent spaces improve sample efficiency, trajectory smoothness, and skill generalization.
- **Tactile Perception: Tactile-VLA** Huang et al. [2025] integrates tactile feedback, enriching action representations and supporting tasks dependent on fine-grained contact dynamics.

By combining 3D perception, memory modules, latent action embeddings, and tactile inputs, these representations enable VLA models to plan and act effectively in physically complex, temporally extended, and multi-modal environments, enhancing scalability and generalization.

5 Challenges and Open Issues

Despite substantial progress in Vision–Language–Action (VLA) research, significant technical and practical hurdles remain, limiting the scalability, reliability, and real-world deployment of current models. These challenges can be summarized as follows:

1. **Data Scarcity and Quality:** High-quality, diverse, and task-relevant datasets remain a bottleneck for VLA systems. While large-scale datasets such as Open X-Embodiment provide multi-robot demonstrations across varied tasks, they are still orders of magnitude smaller than text or image corpora. Collecting dexterous manipulation data with dense action labels is expensive and time-consuming, and existing datasets often exhibit biases in task types, object categories, and environmental configurations. Moreover, many datasets are limited in temporal horizon, fail to capture long-horizon dependencies, or do not include failure cases, restricting the model’s ability to generalize to novel or complex scenarios. Addressing this challenge may involve hybrid pipelines combining real-world demonstrations, egocentric videos, and synthetic data, along with self-supervised or simulation-to-real transfer methods.
2. **Inference Latency and Computational Cost:** Large foundation models such as RT-2 or π_0 provide strong priors for language-conditioned control but are computationally expensive during inference. High model latency limits real-time responsiveness, making deployment on embedded robotic platforms challenging. Additionally, the autoregressive generation of actions, diffusion-based sampling, or flow-based iterative inference further exacerbate runtime overhead. Solutions such as model distillation, quantization, adaptive tokenization, or hierarchical planners that decouple high-level reasoning from low-level execution are active research directions to mitigate these constraints.
3. **3D and Spatial Reasoning:** Although recent models such as 3D-VLA incorporate depth, point clouds, or volumetric representations, efficient reasoning in 3D space remains computationally intensive. Capturing geometric, physical, and relational properties of objects over long horizons, while maintaining differentiable representations suitable for learning, is still an open problem. Moreover, aligning 3D world models with natural language instructions and continuous control signals introduces additional challenges in representation learning and policy grounding, especially for tasks requiring precise spatial manipulation or navigation.
4. **Safety, Robustness, and Verifiability:** End-to-end neural policies are notoriously difficult to verify or certify. Without guarantees on stability, robustness, or constraint satisfaction, deploying VLAs in human-centric environments poses safety risks. Models may fail unpredictably under distributional shifts, novel objects, or dynamic interactions. Techniques for safe exploration, formal verification, uncertainty-aware planning, and risk-sensitive reinforcement learning are critical to mitigate these risks. Incorporating hierarchical architectures, memory-based context, or symbolic reasoning can also improve interpretability and predictability, partially addressing safety concerns.
5. **Long-Horizon and Compositional Tasks:** Many real-world tasks require sequential decision making with long temporal dependencies, conditional branching, and skill composition. Current VLA models struggle with compounding errors and credit assignment over long horizons. Hierarchical decomposition, planning over abstract subgoals, and memory-augmented representations are promising directions, but designing architectures and training strategies that effectively scale to long-horizon tasks remains an open research challenge.
6. **Generalization Across Embodiments and Domains:** VLA models often exhibit limited transferability between different robot morphologies, sensor configurations, or environmental contexts. While multi-embodiment datasets partially address this, aligning perceptual, linguistic, and action representations across heterogeneous platforms is nontrivial. Approaches such as meta-learning, domain adaptation, and modular policy representations are being explored to improve cross-robot and cross-domain generalization.

In summary, addressing these challenges requires a combination of advances in dataset design, model architectures, learning paradigms, and verification techniques. Overcoming these hurdles is essential for scaling VLAs from controlled laboratory settings to robust, autonomous agents in real-world environments.

6 Conclusion

In this survey, we reviewed the emerging landscape of Vision–Language–Action models. We discussed transformer-based baselines such as RT-2 and OpenVLA, as well as generative approaches leveraging flow- and diffusion-based policies like π_0 . These models illustrate the field’s rapid progress toward generalist robotic agents capable of performing diverse tasks in open-world environments. Despite these advances, significant challenges remain, including high inference latency, limited 3D and spatial reasoning, and the need for reinforcement learning to enable continuous self-improvement. Addressing these challenges will be critical for realizing autonomous, robust, and scalable VLA systems in real-world applications.

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